HR Workforce Analytics: an end-to-end dashboard project

Excel, MySQL, Power BI and Tableau on a 100-employee HR dataset

100	67	33	33%
employees	active	left the company	attrition rate

Overview

I took a raw HR dataset of 100 employee records and built the same analysis in four tools: Excel, MySQL, Power BI and Tableau. The goal was to answer one business question: where is the company losing people, and what does the data say about why? This page summarises the dataset, my method, what I found and what I would recommend.

The dataset

Employee_ID, Department, Job_Role, Location	Who the employee is and where they work (8 departments, 8 locations, 29 job roles)
Hire_Date, Exit_Date, Employment_Status	Tenure and outcome: Active, Resigned or Terminated
Annual_Salary	Yearly pay in rupees (INR)
Performance_Rating, Engagement_Score, Satisfaction_Score	Scores on a 1 to 5 scale
Training_Hours	Hours of training per employee

How I built it

1. Clean and explore (Excel). Converted dates, built an Excel Table, added helper columns (tenure, hire year, attrition flag, salary band) and built a KPI dashboard with formulas and charts.

2. Store and query (MySQL). Designed the employees table, loaded all 100 rows and wrote 8 SQL views, one per dashboard chart, using GROUP BY, CASE and date functions.

3. Model and visualise (Power BI). Wrote DAX measures such as Attrition % and Average Tenure, then built an interactive report with slicers.

4. Tell the story (Tableau). Created calculated fields and a colour-coded dashboard with KPI tiles and a mix of chart types.

Key findings

1. A third of the workforce has left. 33 of 100 employees are no longer active: 20 were terminated and 13 resigned.

2. Attrition is concentrated in three departments. Sales loses 75% (6 of 8), Customer Support 55% (6 of 11) and HR 46% (6 of 13). IT and Marketing are lowest at 14% (2 of 14 each).

3. Pune and Coimbatore stand out. Attrition is 46% in Pune and 45% in Coimbatore, against 21% in Bengaluru and 22% in Chennai.

4. Exits are speeding up. 13 employees left in 2025, compared with 4 in 2024. The data already shows 6 exits in 2026.

5. Hiring has slowed. 17 people were hired in each of 2017 and 2019, against 7 in 2025 and 2 in 2026 (a part year).

6. Pay differs by team but does not explain exits on its own. Engineering averages 12.2 lakh a year and Customer Support 4.5 lakh. People who left earned 6.9 lakh on average against 7.4 lakh for active staff, and the correlation between salary and leaving is weak (-0.08).

7. Engagement is low overall and slightly lower for leavers. The average engagement score is 2.83 out of 5: 2.64 for leavers and 2.93 for active staff. IT has the lowest engagement (2.21) yet the lowest attrition.

8. Good performers leave at the same rate as everyone else. 13 of the 39 employees rated 4 or 5 have left (33%). The average tenure at exit is 3.2 years.

What the dashboards show



Nine charts from the dashboard: headcount, attrition by department, status, location, hiring trend, engagement, performance, salary bands and pay by role.

Recommendations

- Run retention reviews in Sales, Customer Support and HR first, since they hold 18 of the 33 exits.
- Check pay bands for support roles: Support Team Leads average 4.2 lakh and Support Executives 4.3 lakh.
- Look into Pune and Coimbatore (manager, workload, role mix) to see what sets them apart.
- Start recording exit reasons and manager IDs. The data shows where people leave, not why.
- Create a retention plan for high performers, who currently leave as often as anyone.

Limitations

This is a sample dataset of 100 records. Department groups hold only 8 to 20 people, so a single exit moves a percentage a lot. The dataset has no exit reasons, so the findings describe patterns, not causes.

Skills shown in this project

- Excel: tables, formulas, pivot tables, charts
- SQL: MySQL schema design, imports, views, aggregation
- Power BI: DAX measures, slicers, report design
- Tableau: calculated fields, dashboards, colour and layout
- Data cleaning, KPI design and data storytelling